

79. What is the total surface area of the given frustum?

(a) $9\pi(2\sqrt{10} + 5)\text{cm}^2$ (b) $9\pi(3\sqrt{10} + 5)\text{cm}^2$
 (c) $9\pi(3\sqrt{10} + 4)\text{cm}^2$ (d) $27\pi(\sqrt{10} + 1)\text{cm}^2$

PREVIOUS YEARS' QUESTIONS

80. What will be the cost to plaster the inner surface of a well 14 m deep and 4 m in diameter at the rate of ₹ 25 per m^2 ? **2012 I**

(a) ₹ 4000 (b) ₹ 4200 (c) ₹ 4400 (d) ₹ 5400

81. What is the length of the uniform wire of diameter 0.4 cm that can be drawn from a solid sphere of radius 9 cm? **2012 I**

(a) 243 m (b) 240 m (c) 60.75 m (d) 60 m

82. The total surface area of a cube is 150 cm^2 . What is its volume? **2012 I**

(a) 64 cm^3 (b) 81 cm^3 (c) 125 cm^3 (d) 160 cm^3

83. If the volume of a cube is 729 cm^3 , then what is the length of its diagonal? **2012 I**

(a) $9\sqrt{2}\text{ cm}$ (b) $9\sqrt{3}\text{ cm}$ (c) 18 cm (d) $18\sqrt{3}\text{ cm}$

84. The curved surface area of a right circular cone of radius 14 cm is 440 cm^2 . What is the slant height of the cone? **2012 I**

(a) 10 cm (b) 11 cm (c) 12 cm (d) 13 cm

85. A large solid metallic cylinder whose radius and height are equal to each other is to be melted and 48 identical solid balls are to be recast from the liquid metal, so formed. What is the ratio of the radius of a ball to the radius of the cylinder?

(a) 1 : 16 (b) 1 : 12 **2012 I**
 (c) 1 : 8 (d) 1 : 4

86. What are the dimensions (length, breadth and height, respectively) of a cuboid with volume 720 cm^3 , surface area 484 cm^2 and the area of the base 72 cm^2 ? **2012 I**

(a) 9, 8 and 10 cm (b) 12, 6 and 10 cm
 (c) 18, 4 and 10 cm (d) 30, 2 and 12 cm

87. If the surface area of a sphere is 616 cm^2 , then what is its volume? **2012 I**

(a) $4312/3\text{ cm}^3$ (b) $4102/3\text{ cm}^3$
 (c) 1257 cm^3 (d) 1023 cm^3

Directions (Q.Nos. 88-89) Read the following information carefully and answer the given questions that follow.

The areas of the ends of a frustum of a cone are P and Q , where $P < Q$ and H is its thickness.

88. What is the difference in radii of the ends of the frustum? **2012 II**

(a) $\frac{\sqrt{Q} - \sqrt{P}}{\sqrt{\pi}}$ (b) $\frac{\sqrt{Q} - \sqrt{P}}{\pi}$
 (c) $\sqrt{Q} - \sqrt{P}$ (d) None of these

89. What is the volume of the frustum? **2012 II**

(a) $3H(P + Q + \sqrt{PQ})$ (b) $H(P + Q + \sqrt{PQ})$
 (c) $H(P + Q + \sqrt{PQ})/3$ (d) $H(P + Q - \sqrt{PQ})/3$

90. Let the largest possible right circular cone and largest possible sphere be fitted into two cubes of same length. If C and S denote the volume of cone and volume of sphere, respectively. Then, which one of the following is correct? **2012 II**

(a) $C = 2S$ (b) $S = 2C$ (c) $C = S$ (d) $C = 3S$

91. 10 circular plates each of thickness 3 cm, are placed one above the other and a hemisphere of radius 6 cm is placed on the top just to cover the cylindrical solid. What is the volume of the solid so formed? **2012 II**

(a) $264\pi\text{ cm}^3$ (b) $252\pi\text{ cm}^3$
 (c) $236\pi\text{ cm}^3$ (d) None of these

92. A right circular metal cone (solid) is 8 cm high and the radius is 2 cm. It is melted and recast into a sphere. What is the radius of the sphere? **2012 II**

(a) 2 cm (b) 3 cm (c) 4 cm (d) 5 cm

93. The volume of a cube is numerically equal to sum of its edges. What is the total surface area?

(a) 12 sq units (b) 36 sq units **2012 II**
 (c) 72 sq units (d) 144 sq units

94. The diameter of base of a right circular cone is 7 cm and slant height is 10 cm, then what is its lateral surface area? **2012 II**

(a) 110 cm^2 (b) 100 cm^2 (c) 70 cm^2 (d) 49 cm^2

95. What is the height of a solid cylinder of radius 5 cm and total surface area is 660 cm^2 ? **2012 II**

(a) 10 cm (b) 12 cm (c) 15 cm (d) 16 cm

96. If the ratio of the diameters of two spheres is 3 : 5, then what is the ratio of their surface areas? **2012 II**

(a) 9 : 25 (b) 9 : 10 (c) 3 : 5 (d) 27 : 125

97. What is the volume of the largest sphere that can be curved out of a cube of edge 3 cm? **2012 II**

(a) $9\pi\text{ cm}^3$ (b) $6\pi\text{ cm}^3$ (c) $4.5\pi\text{ cm}^3$ (d) $3\pi\text{ cm}^3$

98. What is the quantity of cloth required to roll up to form a right circular tent whose base is of radius 12 m and height 5 m? **2013 I**

(a) $40\pi\text{ m}^2$ (b) $60\pi\text{ m}^2$ (c) $78\pi\text{ m}^2$ (d) $156\pi\text{ m}^2$

99. The volume of the material of a hemispherical shell with outer and inner radii 9 cm and 7 cm, respectively is approximately **2013 I**

(a) 808 cm^3 (b) 800 cm^3 (c) 816 cm^3 (d) 824 cm^3

100. The ratio of surface area to diameter of a sphere whose volume is $36\pi\text{ cm}^3$, is **2013 I**

(a) 3π (b) 6π (c) 6 (d) None of these

- 101.** A cylindrical tube open at both ends is made of metal. The internal diameter of the tube is 6 cm and length of the tube is 10 cm. If the thickness of the metal used is 1 cm, then the outer curved surface area of the tube is **✓ 2013 I**
 (a) $140\pi \text{ cm}^2$ (b) $146.5\pi \text{ cm}^2$
 (c) $70\pi \text{ cm}^2$ (d) None of these
- 102.** From a solid wooden right circular cylinder, a right circular cone whose radius and height are same as the radius and height of the cylinder, respectively is carved out. What is the ratio of the volume of the utilised wood to that of the wasted wood? **✓ 2013 I**
 (a) 1 : 2 (b) 2 : 1 (c) 2 : 3 (d) 1 : 3
- 103.** If the heights and the areas of the base of a right circular cone and a pyramid with square base are the same, then they have **✓ 2013 I**
 (a) same volume and same surface area
 (b) same surface area but different volumes
 (c) same volume but different surface areas
 (d) different volumes and different surface areas
- 104.** The volume of a right circular cone of height 3 cm and slant height 5 cm is **✓ 2013 I**
 (a) 49.3 cm^3 (b) 50.3 cm^3
 (c) 52 cm^3 (d) 53 cm^3
- 105.** A bucket is of a height 25 cm. Its top and bottom radii are 20 cm and 10 cm, respectively. Its capacity (in L) is **✓ 2013 I**
 (a) $17.5\pi/3$ (b) 17.5π (c) 20π (d) 25π
- 106.** The height of a cylinder is 15 cm. The lateral surface area is 660 cm^2 . Its volume is **✓ 2013 II**
 (a) 1155 cm^3 (b) 1215 cm^3 (c) 1230 cm^3 (d) 2310 cm^3
- 107.** The diameter of the Moon is approximately one-fourth of the diameter of the Earth. What is the ratio (approximate) of their volumes? **✓ 2013 II**
 (a) 1 : 16 (b) 1 : 64 (c) 1 : 4 (d) 1 : 128
- 108.** A conical cap has the base diameter 24 cm and height 16 cm. What is the cost of painting the surface of the cap at the rate of 70 paise per cm^2 ? **✓ 2013 II**
 (a) ₹ 520 (b) ₹ 524 (c) ₹ 528 (d) ₹ 532
- 109.** What is the whole surface area of a cone of base radius 7 cm and height 24 cm? **✓ 2013 II**
 (a) 654 cm^2 (b) 704 cm^2 (c) 724 cm^2 (d) 964 cm^2
- 110.** A tent is in the form of a right circular cylinder surmounted by a cone. The diameter of the cylinder is 24 m. The height of the cylindrical portion is 11 m, while the vertex of the cone is 16 m above the ground. What is the area of the curved surface for conical portion? **✓ 2013 II**
 (a) $3434/9 \text{ m}^2$ (b) $3431/8 \text{ m}^2$
 (c) $3432/7 \text{ m}^2$ (d) $3234/7 \text{ m}^2$
- 111.** If x is the curved surface area and y is the volume of a right circular cylinder, then which one of the following is correct? **✓ 2014 I**
 (a) Only the ratio of the height to radius of the cylinder is independent of x
 (b) Only the ratio of height to radius of the cylinder is independent of y
 (c) Either (a) or (b)
 (d) Neither (a) nor (b)
- 112.** A cylinder is surmounted by a cone at one end, a hemisphere at the other end. The common radius is 3.5 cm, the height of the cylinder is 6.5 cm and the total height of the structure is 12.8 cm. The volume V of the structure lies between **✓ 2014 I**
 (a) 370 cm^3 and 380 cm^3 (b) 380 cm^3 and 390 cm^3
 (c) 390 cm^3 and 400 cm^3 (d) None of these
- 113.** The diameter of the base of a cone is 6 cm and its altitude is 4 cm. What is the approximate curved surface area of the cone? **✓ 2014 I**
 (a) 45 cm^2 (b) 47 cm^2 (c) 49 cm^2 (d) 51 cm^2
- 114.** A drainage tile is a cylindrical shell 21 cm long. The inside and outside diameters are 4.5 cm and 5.1 cm, respectively. What is the volume of the clay required for the tile? **✓ 2014 I**
 (a) $6.96\pi \text{ cm}^3$ (b) $6.76\pi \text{ cm}^3$
 (c) $5.76\pi \text{ cm}^3$ (d) None of these
- 115.** A cube has each edge 2 cm and a cuboid is 1 cm long, 2 cm wide and 3 cm high. The paint in a certain container is sufficient to paint an area equal to 54 cm^2 . **✓ 2014 I**
 Which one of the following is correct?
 (a) Both cube and cuboid can be painted
 (b) Only cube can be painted
 (c) Only cuboid can be painted
 (d) Neither cube nor cuboid can be painted
- 116.** A cone of radius r cm and height h cm is divided into two parts by drawing a plane through the middle point of its height and parallel to the base. What is the ratio of the volume of the original cone to the volume of the smaller cone? **✓ 2014 I**
 (a) 4 : 1 (b) 8 : 1 (c) 2 : 1 (d) 6 : 1
- 117.** If 64 identical small spheres are made out of big sphere of diameter 8 cm, then what is surface area of each small sphere? **✓ 2014 I**
 (a) $\pi \text{ cm}^2$ (b) $2\pi \text{ cm}^2$ (c) $4\pi \text{ cm}^2$ (d) $8\pi \text{ cm}^2$
- 118.** The dimensions of a field are 15 m by 12 m. A pit 8 m long, 2.5 m wide and 2 m deep is dug in one corner of the field and the earth removed is evenly spread over the remaining area of the field. The level of the field is raised by **✓ 2014 I**
 (a) 15 cm (b) 20 cm (c) 25 cm (d) $\frac{200}{9} \text{ cm}$

- 119.** What is the diameter of the largest circle lying on the surface of a sphere of surface area 616 cm^2 ?

☑ 2014 I

(a) 14 cm (b) 10.5 cm (c) 7 cm (d) 3.5 cm

- 120.** The volume of a hollow cube is $216x^3$. What is the surface area of the largest sphere which can be enclosed in it?

☑ 2014 I

(a) $18\pi x^2$ (b) $27\pi x^2$ (c) $36\pi x^2$ (d) $72\pi x^2$

Directions (Q. Nos. 121-122) Read the following information carefully and answer the given questions that follow.

A right angled triangle having hypotenuse 25 cm and legs in the ratio 3 : 4 is made to revolve about its hypotenuse. ($\pi = 3.14$)

- 121.** What is the volume of the double cone so formed?

☑ 2014 II

(a) 3124 cm^3 (b) 3424 cm^3 (c) 3768 cm^3 (d) 3924 cm^3

- 122.** What is the surface area of the double cone so formed?

☑ 2014 II

(a) 1101.2 cm^2 (b) 1111.4 cm^2 (c) 1310.4 cm^2 (d) 1318.8 cm^2

- 123.** If the side of a cube is increased by 100%, then by what percentage is the surface area of the cube increased?

☑ 2014 II

(a) 150% (b) 200% (c) 300% (d) 400%

- 124.** A cylinder circumscribes a sphere. What is the ratio of volume of the sphere to that of the cylinder?

☑ 2014 II

(a) 2 : 3 (b) 1 : 2 (c) 3 : 4 (d) 3 : 2

Directions (Q. Nos. 125-126) Read the following information carefully and answer the given questions that follow.

A toy is in the form of a cone mounted on the hemisphere with the same radius. The diameter of the base of the conical portion is 12 cm and its height is 8 cm.

- 125.** What is the total surface area of the toy?

☑ 2014 II

(a) $132\pi \text{ cm}^2$ (b) $112\pi \text{ cm}^2$ (c) $96\pi \text{ cm}^2$ (d) $66\pi \text{ cm}^2$

- 126.** What is the volume of the toy?

☑ 2014 II

(a) $180\pi \text{ cm}^3$ (b) $240\pi \text{ cm}^3$ (c) $300\pi \text{ cm}^3$ (d) $320\pi \text{ cm}^3$

- 127.** The areas of the three adjacent faces of a cuboidal box are x , $4x$ and $9x$ sq unit. What is the volume of the box?

☑ 2014 II

(a) $6x^2$ cu units (b) $6x^{3/2}$ cu units
(c) $3x^{3/2}$ cu units (d) $2x^{3/2}$ cu units

- 128.** What is the number of pairs of perpendicular planes in a cuboid?

☑ 2014 II

(a) 4 (b) 8
(c) 12 (d) None of these

- 129.** Consider the following statements in respect of four spheres A, B, C and D having respective radii 6, 8, 10 and 12 cm.

I. The surface area of sphere C is equal to the sum of surface areas of spheres A and B .

II. The volume of sphere D is equal to the sum of volumes of spheres A, B and C .

☑ 2014 II

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 130.** The diameter of a metallic sphere is 6 cm. The sphere is melted and drawn into a wire of uniform circular cross-section. If the length of the wire is 36 m, then what is its radius?

☑ 2014 II

(a) 0.1 cm (b) 0.01 cm (c) 0.001 cm (d) 1.0 cm

- 131.** What is the maximum distance between two points of a cube of side 2 cm?

☑ 2014 II

(a) $\sqrt{3}$ cm (b) $2\sqrt{3}$ cm (c) $4\sqrt{3}$ cm (d) $2\sqrt{2}$ cm

- 132.** The total outer surface area of a right circular cone of height 24 cm with a hemisphere of radius 7 cm upon its base is

☑ 2015 I

(a) $327\pi \text{ cm}^2$ (b) $307\pi \text{ cm}^2$
(c) $293\pi \text{ cm}^2$ (d) $273\pi \text{ cm}^2$

- 133.** A sphere and a cube have same surface area. The ratio of square of their volumes is

☑ 2015 I

(a) 6 : π (b) 5 : π (c) 3 : 5 (d) 1 : 1

- 134.** The radius of a sphere is equal to the radius of the base of a right circular cone, and the volume of the sphere is double the volume of the cone. The ratio of the height of the cone to the radius of its base is

☑ 2015 I

(a) 2 : 1 (b) 1 : 2 (c) 2 : 3 (d) 3 : 2

- 135.** If the radius of a sphere is increased by 10%, then the volume will be increased by

☑ 2015 I

(a) 33.1% (b) 30% (c) 50% (d) 10%

- 136.** If three metallic spheres of radii 6 cm, 8 cm and 10 cm are melted to form a single sphere, then the diameter of the new sphere will be

☑ 2015 I

(a) 12 cm (b) 24 cm (c) 30 cm (d) 36 cm

- 137.** If the height of a right circular cone is increased by 200% and the radius of the base is reduced by 50%, then the volume of the cone

☑ 2015 I

(a) remains unaltered (b) decreases by 25%
(c) increases by 25% (d) increases by 50%

- 138.** A rectangular paper of 44 cm long and 6 cm wide is rolled to form a cylinder of height equal to width of the paper. The radius of the base of the cylinder so rolled is

☑ 2015 I

(a) 3.5 cm (b) 5 cm (c) 7 cm (d) 14 cm

- 139.** The diagonals of three faces of a cuboid are 13, $\sqrt{281}$ and 20 linear units. Then, the total surface area of the cuboid is **☑ 2015 I**
 (a) 650 sq units (b) 658 sq units
 (c) 664 sq units (d) 672 sq units
- 140.** A cylindrical vessel of radius 4 cm contains water. A solid sphere of radius 3 cm is lowered into the water until it is completely immersed. The water level in the vessel will rise by **☑ 2015 I**
 (a) 1.5 cm (b) 2 cm (c) 2.25 cm (d) 4.5 cm
- 141.** A rectangular block of wood having dimensions $3\text{m} \times 2\text{m} \times 1.75\text{m}$ has to be painted on all its faces. The layer of paint must be 0.1 mm thick. Paint comes in cubical boxes having their edges equal to 10 cm. The minimum number of boxes of paint to be purchased is **☑ 2015 I**
 (a) 5 (b) 4 (c) 3 (d) 2
- 142.** If the surface area of a cube is 13254 cm^2 , then the length of its diagonal is **☑ 2015 II**
 (a) $44\sqrt{2}\text{ cm}$ (b) $44\sqrt{3}\text{ cm}$ (c) $47\sqrt{2}\text{ cm}$ (d) $47\sqrt{3}\text{ cm}$
- 143.** A water tank, open at the top, is hemispherical at the bottom and cylindrical above it. The radius is 12 m and the capacity is $3312\pi\text{ m}^3$. The ratio of the surface areas of the spherical and cylindrical portions, is **☑ 2015 II**
 (a) 3 : 5 (b) 4 : 5 (c) 1 : 1 (d) 6 : 5
- 144.** A large water tank has the shape of a cube. If 128 m^3 of water is pumped out, the water level goes down by 2m. Then, the maximum capacity of the tank is **☑ 2015 II**
 (a) 512 m^3 (b) 480 m^3 (c) 324 m^3 (d) 256 m^3
- 145.** The areas of three mutually perpendicular faces of a cuboid are x , y , z . If V is the volume, then xyz is equal to **☑ 2015 II**
 (a) V (b) V^2 (c) $2V$ (d) $2V^2$
- 146.** How many right angled triangles can be formed by joining the vertices of a cuboid? **☑ 2015 II**
 (a) 24 (b) 28 (c) 32 (d) None of these
- 147.** Three rectangles R_1 , R_2 and R_3 have the same area. Their lengths x_1 , x_2 and x_3 respectively are such that $x_1 < x_2 < x_3$. If V_1 , V_2 and V_3 are the volumes of the cylinders formed from the rectangles R_1 , R_2 and R_3 respectively by joining the parallel sides along the breadth, then which one of the following is correct? **☑ 2015 II**
 (a) $V_3 < V_2 < V_1$ (b) $V_1 < V_3 < V_2$
 (c) $V_1 < V_2 < V_3$ (d) $V_3 < V_1 < V_2$
- 148.** Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive Y-axis. The cone is filled with water up to half of its height. The volume of water is **☑ 2015 II**
 (a) $\frac{V}{8}$ (b) $\frac{V}{6}$ (c) $\frac{V}{3}$ (d) $\frac{V}{2}$
- 149.** 30 metallic cylinders of same size are melted and cast in the form of cones having the same radius and height as those of the cylinders. Consider the following statements
 I. A maximum of 90 cones will be obtained.
 II. The curved surface of the cylinder can be flattened in the shape of a rectangle but the curved surface of the cone when flattened has the shape of triangle.
 Which one of the following statement(s) is/are correct in respect of the above? **☑ 2015 II**
 (a) Both statement I and statement II are correct and statement II is the correct explanation of statement I
 (b) Both statement I and statement II are correct and statement II is not the correct explanation of statement I
 (c) statement I is correct but statement II is not correct
 (d) statement I is not correct but statement II is correct
- 150.** Water is filled in a container in such a manner that its volume doubles every 5 min. If it takes 30 min for the container to be full, in how much time will it be one-fourth full? **☑ 2015 II**
 (a) 7.5 min (b) 15 min (c) 20 min (d) 17.5 min
- 151.** Consider the following statements:
 I. If the height of a cylinder is doubled, the area of the curved surface is doubled.
 II. If the radius of a hemispherical solid is doubled, its total surface area becomes fourfold.
 Which of the statement(s) given above is/are correct? **☑ 2015 II**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 152.** From the solid gold in the form of a cube of side length 1 cm, spherical solid balls each having the surface area $\pi^{1/3}\text{ cm}^2$ are to be made. Assuming that there is no loss of the material in process of making the balls, the maximum number of balls made will be **☑ 2015 II**
 (a) 3 (b) 4 (c) 6 (d) 9
- 153.** A hollow cylindrical drum has internal diameter of 30 cm and a height of 1 m. What is the maximum number of cylindrical boxes of diameter 10 cm and height 10 cm each that can be packed in the drum? **☑ 2015 II**
 (a) 60 (b) 70 (c) 80 (d) 90
- 154.** A pipe with square cross-section is supplying water to a cistern which was initially empty. The area of cross-section is 4 cm^2 and the nozzle velocity of water is 40 m/s . The dimensions of the cistern $10\text{ m} \times 8\text{ m} \times 6\text{ m}$. Then, the cistern will be full in **☑ 2015 II**
 (a) 9.5 h (b) 9 h
 (c) 8 h and 20 min (d) 8 h

ANSWERS

1	c	2	b	3	b	4	c	5	d	6	b	7	c	8	a	9	c	10	c
11	c	12	c	13	b	14	a	15	b	16	c	17	d	18	a	19	b	20	a
21	d	22	d	23	b	24	b	25	c	26	b	27	b	28	b	29	b	30	c
31	c	32	c	33	b	34	b	35	b	36	b	37	d	38	a	39	b	40	c
41	c	42	c	43	a	44	b	45	a	46	d	47	b	48	c	49	a	50	b
51	b	52	b	53	c	54	b	55	d	56	d	57	d	58	d	59	b	60	d
61	a	62	c	63	a	64	d	65	b	66	b	67	b	68	a	69	c	70	c
71	d	72	d	73	c	74	c	75	c	76	d	77	a	78	d	79	b	80	c
81	a	82	c	83	b	84	a	85	d	86	a	87	a	88	a	89	c	90	b
91	d	92	a	93	c	94	a	95	d	96	a	97	c	98	d	99	a	100	b
101	d	102	a	103	c	104	b	105	a	106	d	107	b	108	c	109	b	110	c
111	d	112	a	113	b	114	d	115	a	116	b	117	c	118	c	119	a	120	c
121	c	122	d	123	c	124	a	125	a	126	b	127	b	128	c	129	c	130	a
131	b	132	d	133	a	134	a	135	a	136	b	137	b	138	c	139	c	140	c
141	c	142	d	143	b	144	a	145	b	146	a	147	a	148	a	149	b	150	c
151	c	152	c	153	d	154	c												

HINTS AND SOLUTIONS

1. (c) Let 'a' be the edge of three equal cubes. Surface area of a single cube
 $= 6 (\text{Side})^2 = 6a^2$

$\therefore$ Sum of the surface areas of the three cubes is $= 3(6a^2) = 18a^2$

Length of resulting cuboid $= 3a$

Breadth of resulting cuboid $= a$

Height of resulting cuboid $= a$

$\therefore$ Total surface area of cuboid

$$= 2(lb + bh + hl)$$

$$= 2(3a^2 + a^2 + 3a^2) = 14a^2$$

$$\text{Required ratio} = \frac{14a^2}{18a^2} = \frac{7}{9} \text{ i.e. } 7:9$$

2. (b) Here, $x = 12$,

According to the formula, percentage increase in volume

$$= \left[\left(1 + \frac{x}{100} \right)^3 - 1 \right] \times 100\%$$

$$= \left[\left(1 + \frac{12}{100} \right)^3 - 1 \right] \times 100\%$$

$$= \left[\left(1 + \frac{3}{25} \right)^3 - 1 \right] \times 100\%$$

$$= \left[\left(\frac{28}{25} \right)^3 - 1 \right] \times 100\%$$

$$= \left[\frac{21952}{15625} - 1 \right] \times 100\% = [1.405 - 1] \times 100\%$$

$$= 0.405 \times 100\% = 40.5\%$$

3. (b) Given, diameter of the base of the cylinder $= 21$ cm

$$\therefore \text{Radius} = \frac{21}{2} \text{ cm}$$

$$\text{As, curved surface area} = 2\pi rh = 1320$$

$$\Rightarrow 2 \times \frac{22}{7} \times \frac{21}{2} \times h = 1320$$

$$\therefore h = \frac{1320}{22 \times 3} = 20 \text{ cm}$$

4. (c) 1 cm^3 of water weighs 1 g.

$$\therefore \text{Volume of } 154 \text{ g of water} = 154 \text{ cm}^3$$

$$\therefore \text{Volume of cylinder} = \pi r^2 h = 154 \text{ cm}^3$$

$$\Rightarrow \frac{22}{7} (3.5)^2 \times h = 154$$

$$\Rightarrow \frac{77h}{2} = 154$$

$$\Rightarrow h = \frac{154 \times 2}{77} = 4 \text{ cm}$$

5. (d) Here, $2\pi rh = 264 \text{ m}^2$

$$\text{and } \pi r^2 h = 924 \text{ m}^3 \quad [\text{given}]$$

$$\therefore \frac{\pi r^2 h}{2\pi rh} = \frac{924}{264}$$

$$\Rightarrow \frac{r}{2} = \frac{7}{2} \Rightarrow r = 7 \text{ m}$$

$$\therefore \text{Diameter of pillar}$$

$$= r \times 2 = 7 \times 2 = 14 \text{ m}$$

6. (b) Given, $r = 7$ m, $h = 24$ m

Slant height of cone,

$$= \sqrt{r^2 + h^2} = \sqrt{7^2 + 24^2}$$

$$= \sqrt{576 + 49} = \sqrt{625} = 25 \text{ m}$$

Curved surface area $= \pi rl$

$$= \frac{22}{7} \times 7 \times 25 = 550 \text{ m}^2$$

Width of cloth $= 50$ m

$$\text{Length required} = \frac{550}{50} = 11 \text{ m}$$

7. (c) Let radius and height of a right circular cone be $5x$ and $12x$ respectively.

$$\therefore r = 5x \text{ and } h = 12x$$

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h = 2512$$

$$\Rightarrow \frac{1}{3} \times 3.14 \times (5x)^2 \times 12x = 2512$$

$$\Rightarrow x^3 = \frac{2512}{314} = 8$$

$$\Rightarrow x = 2$$

$$\therefore \text{Radius} = 10 \text{ cm, height} = 24 \text{ cm}$$

$\therefore$ Slant height,

$$l = \sqrt{r^2 + h^2} = \sqrt{10^2 + 24^2}$$

$$= \sqrt{676} = 26 \text{ cm}$$